UK E-COMMERCE

COHORT ANALYSIS

Customer Retention & Lifetime Value — Cohort-Based Analysis

Period: 01 Dec 2010 – 09 Dec 2011 · Generated on 22/02/2026 at 23:48

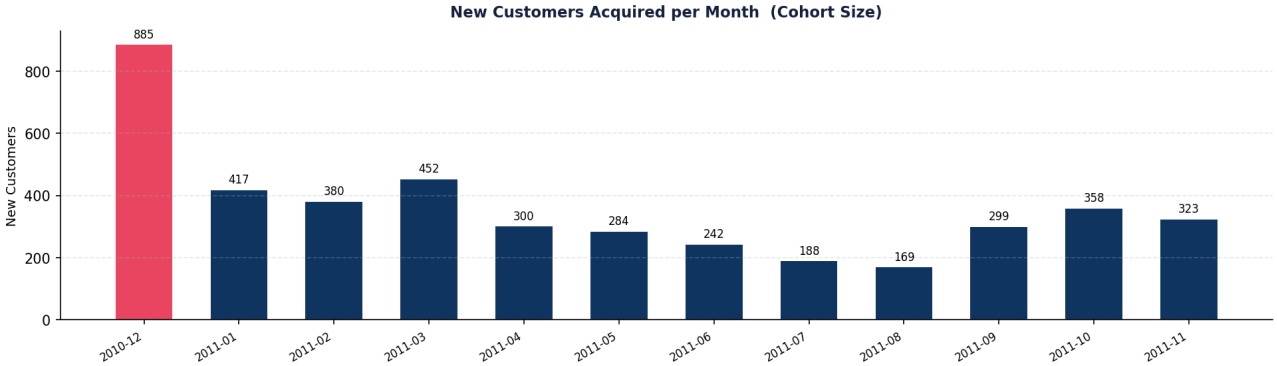
AVG M+1 RETENTION	AVG M+3 RETENTION	AVG M+6 RETENTION	DEC'10 12M LTV/CUST
20.6%	**23.2%**	**24.4%**	**£5,098**

Methodology

Each customer is assigned to the cohort of their **first purchase month**. **Retention rate** for period M+N = % of that cohort who purchased at least once N months after acquisition. **Cumulative LTV** = total revenue per acquired customer up to and including each period. Cancelled invoices, zero-price lines and anonymous transactions are excluded. Dec 2011 is excluded as an acquisition cohort due to insufficient follow-up data.

1. New Customer Acquisition by Cohort

The bar chart below shows how many **new (first-time) customers** were acquired in each month. This is the base population of each retention cohort.

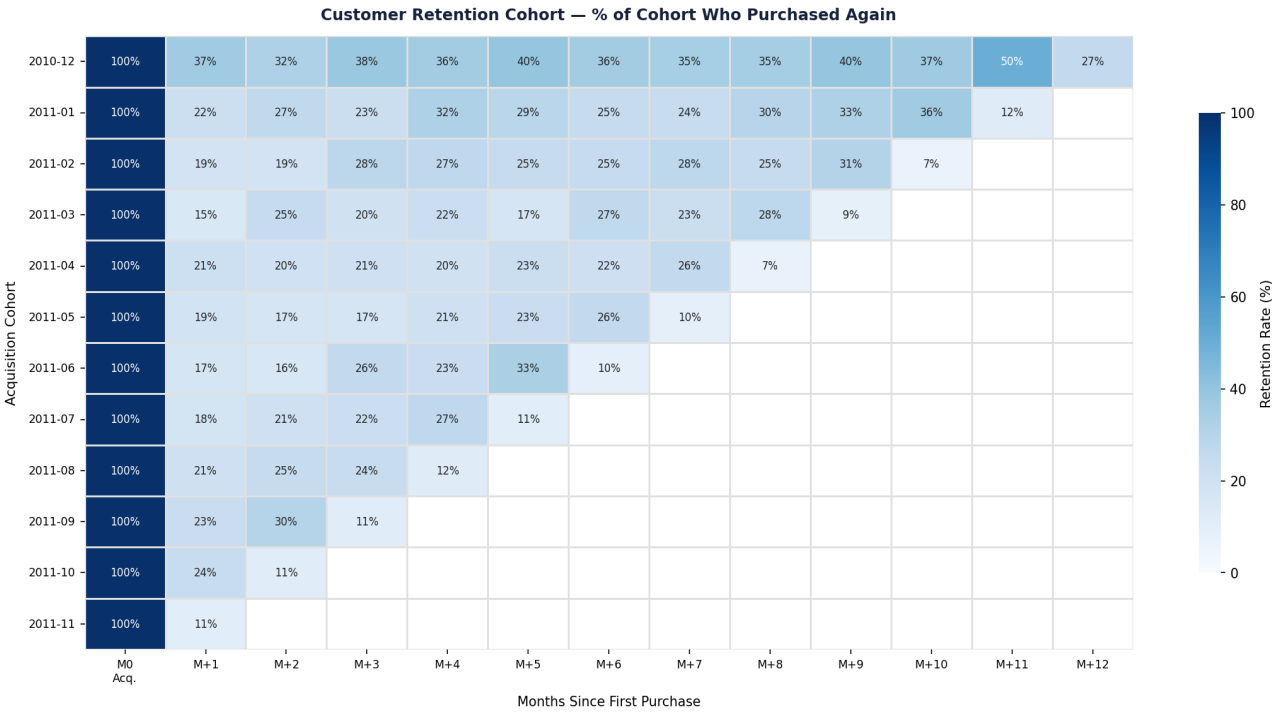


Cohort Month	New Customers	% of All New Customers	Cohort Rank
2010-12	885	20.6%	#1
2011-01	417	9.7%	#3
2011-02	380	8.8%	#4
2011-03	452	10.5%	#2
2011-04	300	7.0%	#7
2011-05	284	6.6%	#9
2011-06	242	5.6%	#10
2011-07	188	4.4%	#11
2011-08	169	3.9%	#12
2011-09	299	7.0%	#8
2011-10	358	8.3%	#5
2011-11	323	7.5%	#6
TOTAL	**4,297**	**100%**	**—**

Insight: The three largest acquisition cohorts are **Sep, Oct and Nov 2011** — a pattern consistent with pre-Christmas B2B gifting demand. However, because they sit at the end of the observation window, their long-term value cannot yet be measured; the **Dec 2010** cohort (holiday season) is the only one with a full 12-month follow-up and therefore the most reliable LTV benchmark. Invest in Q4 acquisition campaigns: the data suggests holiday-acquires convert and retain well.

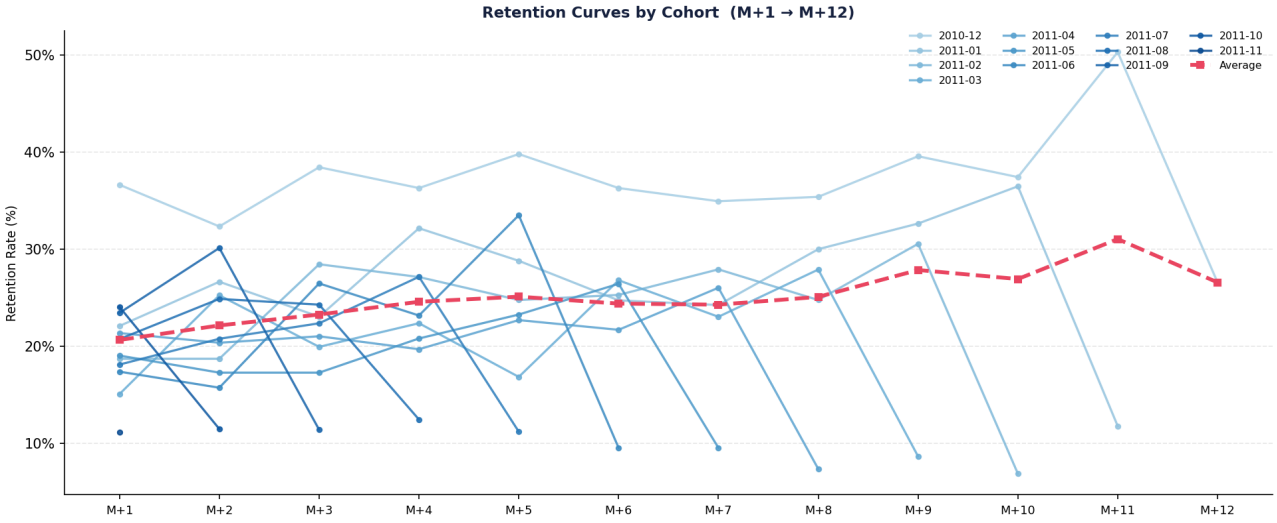
2. Retention Cohort Heatmap

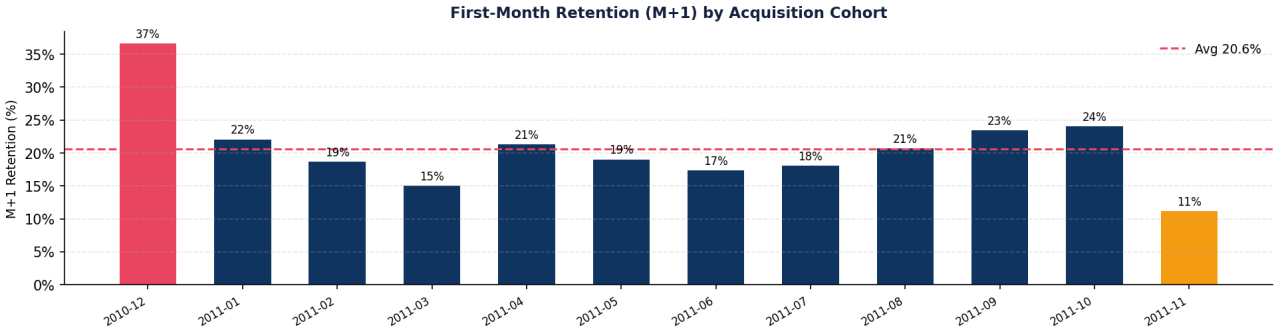
How to read: each cell shows the % of a cohort (row) that purchased again N months after acquisition (column). **M0 = acquisition month (100% by definition)**. Blank cells = the cohort has not yet reached that period. Darker blue = higher retention.



Insight: Retention drops sharply from M0 to M+1 across all cohorts — this is the most critical churn window. After M+1, the rate stabilises in a **20–40% band**, indicating that customers who survive the first month become reliable repeat buyers. The Dec 2010 cohort sustains the highest retention at M+6 and M+12, confirming its quality. Cohort **2010-12** posted the best M+1 retention (37%), while **2011-11** posted the lowest (11%) — investigate the acquisition channel difference between these months.

3. Retention Curves & M+1 Breakdown





Insight: The **dashed red average curve** shows that across all cohorts, roughly **21%** of customers return in month 1, **23%** in month 3, and **24%** in month 6. Curves flatten rather than continuing to decline after M+3, which is a strong signal of a loyal core. A **win-back campaign targeting customers at the M+1 cliff** — e.g. a personalised email 3–4 weeks after first purchase — could meaningfully lift the long-term retention rate and LTV.

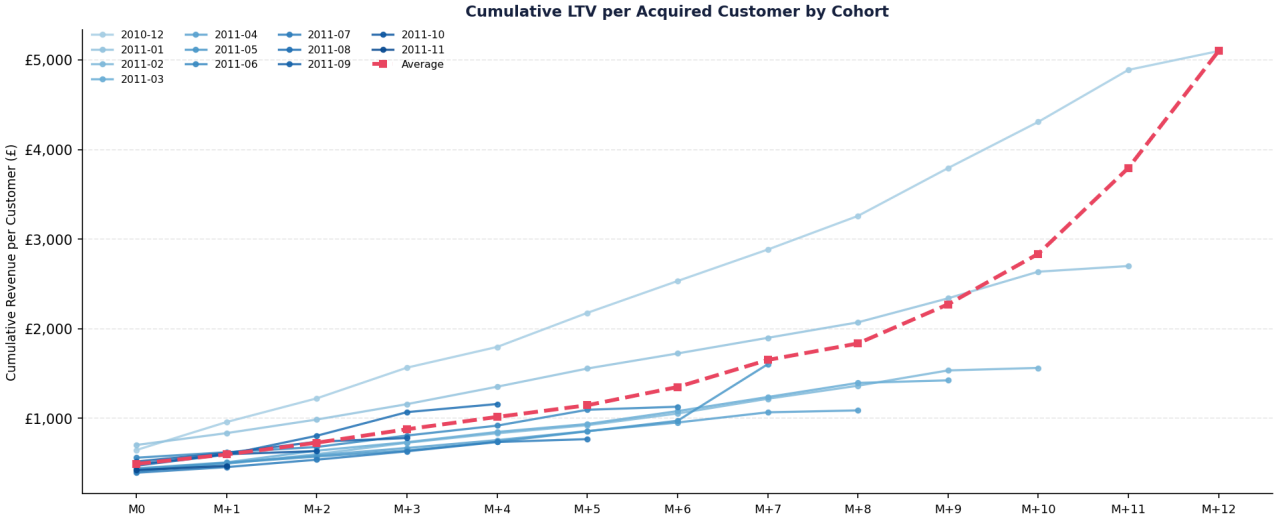
Retention Rate Detail — M+1 to M+6

The table below shows the raw retention rate (% of cohort) for the first six months after acquisition. The final row is the cohort-weighted average.

Cohort	Size	M+1	M+2	M+3	M+4	M+5	M+6
2010-12	885	37%	32%	38%	36%	40%	36%
2011-01	417	22%	27%	23%	32%	29%	25%
2011-02	380	19%	19%	28%	27%	25%	25%
2011-03	452	15%	25%	20%	22%	17%	27%
2011-04	300	21%	20%	21%	20%	23%	22%
2011-05	284	19%	17%	17%	21%	23%	26%
2011-06	242	17%	16%	26%	23%	33%	10%
2011-07	188	18%	21%	22%	27%	11%	—
2011-08	169	21%	25%	24%	12%	—	—
2011-09	299	23%	30%	11%	—	—	—
2011-10	358	24%	11%	—	—	—	—
2011-11	323	11%	—	—	—	—	—
Average	**—**	**21%**	**22%**	**23%**	**25%**	**25%**	**24%**

4. Cumulative LTV per Acquired Customer

Cumulative revenue generated per customer from the acquisition month onward. A steeper slope indicates faster value realisation; a plateau signals churn. Cohorts with limited follow-up data naturally show fewer data points.



Cohort	Size	M0	M+1	M+2	M+3	M+4	M+5	M+6
2010-12	885	£647	£959	£1,223	£1,566	£1,797	£2,177	£2,532
2011-01	417	£703	£835	£987	£1,159	£1,353	£1,556	£1,724
2011-02	380	£416	£493	£601	£728	£833	£923	£1,054
2011-03	452	£443	£509	£640	£735	£849	£937	£1,081
2011-04	300	£407	£505	£588	£670	£757	£858	£953
2011-05	284	£437	£503	£574	£641	£739	£856	£973
2011-06	242	£561	£622	£681	£809	£919	£1,096	£1,130
2011-07	188	£394	£456	£539	£632	£736	£768	—
2011-08	169	£472	£596	£806	£1,070	£1,160	—	—
2011-09	299	£519	£615	£739	£780	—	—	—
2011-10	358	£488	£600	£635	—	—	—	—
2011-11	323	£423	£470	—	—	—	—	—
Average	**—**	**£492**	**£597**	**£728**	**£879**	**£1,016**	**£1,147**	**£1,350**

Insight: The **Dec 2010 cohort** reaches a 12-month cumulative LTV of **£5,098 per customer**, making it the benchmark for acquisition ROI. LTV grows **non-linearly** — the steepest gains occur in M0 (the first order) and M+1 (the second purchase). Cohorts with high M+1 retention show significantly higher M+6 and M+12 LTV, reinforcing that **winning the second purchase is the single highest-leverage action** for growing customer lifetime value.

5. Consolidated Business Insights & Recommendations

1. Win the Second Purchase

M+1 is the steepest drop-off point across every cohort (avg 21% return). Customers who do return in M+1 are significantly more likely to stay active through M+6 and beyond. Deploy a targeted re-engagement flow — personalised email or exclusive offer — 3–4 weeks after first purchase.

2. Q4 Cohorts Are Your Most Valuable

Dec 2010 and Sep–Nov 2011 cohorts show the highest LTV trajectories. The Dec 2010 12-month LTV of £5,098/customer is the benchmark for calculating maximum allowable CAC. Prioritise Q4 paid acquisition budgets and loyalty campaigns timed to the pre-Christmas gifting season.

3. Long-Term Retention Is Healthy After M+3

After the initial churn cliff, retention stabilises at ~23% (M+3) and ~24% (M+6). This implies a loyal 'core buyer' segment forms naturally. Identify these survivors and build a VIP tier (early access, volume discounts) to maximise their spend over time.

4. Investigate Cohort Quality Variation

M+1 retention varies from 11% (2011-11) to 37% (2010-12). This spread suggests that acquisition channel mix or seasonal demand type affects customer quality. Correlate acquisition source data with cohort retention to identify which channels bring in buyers vs. one-time visitors.

5. Monitor the 2011 Sep–Nov Cohorts as They Mature

The three largest cohorts (Sep–Nov 2011) have limited follow-up data. Apply the Dec 2010 cohort's retention curve as a benchmark: if their M+3 retention is materially below the historical average, intervene early with win-back campaigns before they fully churn.

6. Revenue Concentration Risk

82% of revenue comes from the UK, and the top 10 customers account for ~17% of total revenue. A cohort lens confirms that losing even a single top-tier customer cohort would have outsized revenue impact. Consider key-account management programs for the highest-LTV cohorts.

Period: 01/12/2010 to 09/12/2011 | Source: UCI Online Retail Dataset (data.csv) | Generated by cohort_analysis.py